

Ideas Parked — Not Yet Merged (v18)

Staging document; each entry marks its eventual destination

The full working-out of Gap 3 (ideas_v1: “reducing a path by merging close actors”), reached over several turns and recorded here as one connected argument. Its ending differs from Gap 1’s and Gap 2’s: this is not a resolution to merge into Paper 1, but a determination that most of the question was never Paper 1’s to answer. Paper 1 has already been narrowed accordingly (Pattern_Arithmetic_v15_22), keeping only the mechanical guarantee; everything below is destined for Inference instead.

1 Where the discussion started, and the correction made along the way

The first pass treated “direction” in Gap 3’s own notation ($\chi' = \arg(z)$) as $\hat{n}$, and invoked Gap 1’s refusal of Mix across differing $\hat{n}$ against it. That was wrong, and worth recording as a correction rather than quietly dropped: χ is phase at a *shared* direction, the same coordinate Mix’s own flagship example combines — red ($\chi = 0$) plus green ($\chi = 120$) gives yellow ($\chi = 60$), verified directly, Mix’s validated central case, not a refused one. So combining different χ values at one shared $\hat{n}$ was never the problem. The real issue, once that confusion was cleared, turned out to be order-blindness: checked directly, three actors merged in one sequence and the same three merged in reverse give identical magnitude and angle, since addition is commutative regardless of order.

2 Where a path’s commitment to order actually comes from

Checked against the paper’s own words: an observed sequence arrives already ordered from outside the system; Bundle’s first act is to discard that order deliberately ($A \uplus A = A$, order-blind by construction); T ’s actors are then “taken from the bowl or from the tape” — a choice made when a walk is built, not generated by any axiom. The sharp consequence: choosing to route data through T rather than leaving it in a bowl *is itself* the declaration that order matters for that data. Nothing mints that commitment internally; the modeler asserts it once, by which operator they reach for.

3 The temperature example, and why it was never a counterexample

10,000 readings in a minute, reduced to one average: checked against the above, this is Bowl $\rightarrow$ Mix, never routed through T at all. The decision “one reading per minute suffices” *is* the declaration that intra-minute order is noise. Order-blindness only bites data that was already committed to T — asserting, by that earlier choice, that the trajectory itself was part of the answer. The same raw readings support at least three different questions, and which representation is correct depends on which question is being asked, not on the data itself:

- *What was the average?* — order within the window is noise. Bowl, Mix, one point. Correct as given.
- *Did it spike and return?* — the trajectory is the answer; collapsing to one point erases the one thing being asked for.
- *Is the broader trend rising or falling?* — now the sequence of per-window averages is itself ordered, and that order matters again, one level up.

4 Three techniques, verified as answering three different questions, not one technique with three names

Average (one point, possibly manufactured). Checked directly: a spike path ($10 \rightarrow 40 \rightarrow 70 \rightarrow 40 \rightarrow 10$) and a flat path sitting at 40 throughout produce the identical single output (33.79 averaged, or 40 for the real closest point either way) — one number cannot distinguish “it spiked” from “it never moved,” whether that number is manufactured or picked from the real data. Realness was never the issue; count was.

Multiple centroids (several points, grouped by value, blind to position). Checked directly: a forward spike sequence and its reverse — same multiset of values, opposite order — produce *identical* 2-means centroids ($\{22, 70\}$ either way). Clustering answers “what regimes occurred” correctly and efficiently; it cannot answer “in what order,” by construction, since position was never an input to the grouping.

Shape-preserving reduction (several real points, kept at their original positions). Keeping start/peak/end from the same spike — three real values, in original sequence — lets a reader see the rise and the return. This is not a new technique: it is exactly what line-simplification algorithms (Douglas–Peucker, canonically) already do for ordinary paths, parameterized by a tolerance ε that bounds how far the simplified path may deviate from the original. That ε is the drift bound `ideas_v1`’s own Gap 3 text asked for, under a different name, already solved elsewhere, not yet ported here.

5 What stays in Paper 1, and why (already executed)

Per the algebra’s own existing principle — a reduction is a second, optional reading, not a replacement, the same promise Collapse already keeps for its own sources — Paper 1 states only that any reduction must not consume what it reduces. It does not choose among the three techniques above, because that choice depends on purpose, and purpose is Inference’s business, not the algebra’s. This has already been written into `Pattern Arithmetic_v15.22.tex`, narrowing the former Future-Work paragraph to exactly this mechanical content and redirecting the rest here.

6 What is destined for Inference, specifically

- The technique-selection question itself: average versus centroid versus shape-preservation, chosen by what the reduction needs to answer, as argued throughout this entry.

- The stated distance and drift bound, meaningful only once a technique and a purpose are both fixed — not general algebra facts, but parameters of a specific reference-atom or classification need.
- The one genuinely unbuilt piece, narrowed considerably by this discussion: adapting Douglas–Peucker’s planar deviation measure to this geometry specifically. The original algorithm measures straight-line distance from a chord between two kept points; here, “the chord between two kept actors” needs to mean something stated in terms of T ’s own rotations — plausibly geodesic deviation from the great-circle arc the two kept actors would define, not planar distance. This adaptation, not the general principle, is the remaining work.

7 Status

Gap 3 is closed in the sense that matters for Paper 1: the paper no longer owes an answer it was never positioned to give, and says so directly rather than leaving an unresolved paragraph sitting in its own Future Work. The technique-selection question is fully specified here — three distinct tools, each verified against the question it actually answers, not three names for one idea — and ready for Inference to pick up, pending the one adaptation named above.